

# JUNLI GONG

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## RESEARCH INTERESTS

Efficient, scalable systems for multimodal and Mixture-of-Experts models, with a focus on LLM inference, communication-aware distributed execution, and hardware-aware acceleration.

## EDUCATION

- Northeastern University** Sep 2025 – Dec 2026 (expected)  
*Master of Science in Computer Science, Khoury College of Computer Sciences* Seattle, WA, USA
- **GPA:** 3.89/4.00; **Seattle Campus Scholarship** (merit-based)
  - **Relevant Courses:** Machine Learning, Scalable Distributed Systems, Reinforcement Learning, Algorithms
- Beijing Normal-Hong Kong Baptist University (BNBU; formerly UIC)** Sep 2021 – Jun 2025  
*Bachelor of Science (Honours) in Computer Science and Technology* Zhuhai, China
- **Honors:** President's Honour Roll; Dean's List (4 semesters)

## PUBLICATIONS

\* denotes equal contribution.

### Peer-reviewed

- [1] Guo, Y., **Gong, J.**, Cai, H., Cheung, Y.-m., & Su, W. (2026). Can Segmentation Models Understand the World? Proactive Affordance Reasoning via Vision-Grounded Chain-of-Thought. *EMNLP 2026* (Main Conference, **Oral**).
- [2] Guo, Y.\*, **Gong, J.\***, Wang, W., Cai, H., Cheung, Y.-m., & Su, W. (2026). PEAM: Parametric Embodied Agent Memory for Continual Skill Learning via Internalization of Experience in Minecraft. *EMNLP 2026* (Findings).
- [3] Guo, Y., **Gong, J.**, Dong, W., Cheung, Y.-m., & Su, W. (2026). Adding Thermal Awareness to Visual Systems in Real-Time via Distilled Diffusion Models. *CVPR 2026 Workshop* (World Models Meet Active Sensing).
- [4] Zheng, C.\*, **Gong, J.\***, Guo, J., Tang, Z., & Wang, T. (2025). Enhancing Federated Learning in IoT Through Dynamic Spectral Clustering. *WASA 2025*.
- [5] Fei, C., Xia, N., Tsai, P.-W., Lu, Y., Pan, X., & **Gong, J.** (2024). An Effective Feature Selection Algorithm for Machine Learning-based Malicious Traffic Detection. *AsiaJCIS 2024*.

### Under Review

- [6] Guo, Y.\*, **Gong, J.\***, Lu, Y., Xu, X., Cheung, Y.-m., & Su, W. (2026). Bringing Multimodal Large Language Models to Infrared-Visible Image Fusion Quality Assessment. Preprint. arXiv:2605.06969.
- [7] Guo, Y., **Gong, J.**, Cheung, Y.-m., & Su, W. (2026). LumiVideo: An Intelligent Agentic System for Video Color Grading. Preprint. arXiv:2604.02409.
- [8] Guo, Y., **Gong, J.**, Cheung, Y.-m., & Su, W. (2024). Fuse4Seg: Image Fusion for Multi-Modal Medical Segmentation via Bi-level Optimization. Preprint. arXiv:2409.10328.

## RESEARCH EXPERIENCE

- Graduate Researcher** Jan 2026 – Present  
*Khoury College of Computer Sciences, Northeastern University* Advisor: Prof. Yifan Hu
- Investigating speculative expert pre-dispatch for Mixture-of-Experts inference: lightweight linear probes predict expert routing from pre-attention hidden states so that the expert-parallel AllToAll can overlap with attention, while a verify-and-fallback step keeps outputs identical to the serial path.
  - Preliminary end-to-end results (DeepSeek-V2-Lite, prefill, 2×A100, EP=2, unfused SGLang prototype): gains appear once per-layer AllToAll latency exceeds the break-even point between 2 and 5 ms (near EP=4 under a fitted scaling model), reaching −22% median and P99 TTFT at an injected 5 ms delay that emulates PCIe/cross-node links. Expert-level probe metrics, higher-EP trace replay, and decode-phase (TPOT) evaluation are in progress toward a workshop/arXiv submission.
- Research Assistant** Sep 2024 – Present  
*Department of Computer Science, Beijing Normal-Hong Kong Baptist University* Supervisor: Prof. Weifeng Su
- Drove the experimental research for six papers on multimodal perception and embodied AI [1–3, 6–8]: built the training and evaluation pipelines, designed and ran the ablation studies, and wrote the experiments sections.
  - Co-first-authored PEAM [2] (Findings of EMNLP 2026): designed the failure-correction contrastive training objective that internalizes corrected trajectories into per-category MoE-LoRA skill adapters, and implemented and evaluated the full framework in Minecraft, reaching 69.7% success on a held-out long-horizon task suite—+15.2 points over VOYAGER and +9.1 over Optimus-1—with 6.8× fewer tokens per task.

- Co-first-authored FuScore [6]: designed the Thurstone-based within- and cross-source ranking objectives and built the MLLM-based infrared–visible fusion quality assessor around them, from zero-shot scoring to fine-tuning with soft-label distributional supervision; it outperforms every external baseline on all five rank metrics on EVAFusion (+0.094 SRCC over the strongest), with the soft-label/tripartite-loss recipe alone adding +0.111 SRCC on the same backbone.
- For SegWorld [1] (EMNLP 2026), implemented the two-pass reasoning pipeline (instruction-free proactive observation, then instruction-conditioned visual chain-of-thought) on Sa2VA-Qwen2.5-VL-7B and ran the Intent2Part evaluation (intent-level mIoU 0.514 zero-shot, 0.612 with intent supervision, above all fine-tuned baselines); for FusionProxy [3], built the distilled-diffusion real-time thermal–visible fusion module (84 FPS on an RTX 4070 in FP16) and its closed-loop CARLA driving evaluation.

**Research Intern**

Jul 2025 – Aug 2025

*Pazhou Lab / School of Computer Science and Engineering, South China University of Technology* Supervisor: Prof. Lu Lu

- Developed a custom FP16 Tensor Core HGEMM kernel in CUDA for Transformer-shaped workloads on NVIDIA A800, with coalesced global-memory access and double-buffered software pipelining (+6–9%).
- Benchmarked against cuBLAS on five BERT/LLaMA-derived shapes: 1.33× faster on the bandwidth-bound attention-score shape ( $1024 \times 1024 \times 64$ ) and 11–39% of cuBLAS on the four compute-bound shapes; traced the gap to shared-memory bandwidth and pipeline depth, pointing to CUTLASS-style multi-stage pipelining (overlapping shared-memory-to-register loads with MMA execution) and shape-adaptive kernel selection as the next steps.

**Undergraduate Researcher**

Jan 2024 – Jun 2025

*Advanced Institute of Natural Sciences, Beijing Normal University, Zhuhai*

Supervisor: Prof. Jianxiong Guo

- Co-first-authored a WASA 2025 paper [4] on clustered federated learning for non-IID IoT data: designed DSCFL, a dynamic spectral-clustering scheme that regroups clients by data similarity during training, and implemented it in Python together with a dynamic resource-allocation strategy.
- DSCFL beats FedAvg and CFL (Sattler et al., TNNLS'20) on EMNIST, CIFAR-10, and FashionMNIST in all six settings—up to +7.9 points over FedAvg and +6.4 over CFL with 80 clients—and recovers the correct client clusters by round 5 versus round 62 for CFL.

**Research Assistant**

Jan 2024 – Jun 2024

*School of Computer and Electronic Information, Nanjing Normal University (remote)*

Supervisor: Dr. Nian Xia

- Collected and preprocessed traffic datasets (TII-SSRC-23, UNSW-NB15), trained SVM and decision-tree detectors with EFS-CST feature selection, and wrote the data-collection and evaluation sections of the AsiaJCIS 2024 paper [5].

**INDUSTRY EXPERIENCE****iFLYTEK Co., Ltd.**

Jul 2026 – Aug 2026

*AI Infra Intern**Hefei, China*

- Developed and debugged an automated evaluation platform for adapting speech models (ASR/TTS) to heterogeneous domestic AI accelerators (Ascend NPU, DCU, MLU, MUSA): resolved 30 engineering issues, redesigned task-liveness detection, and built a result-integrity validator that flagged all 99 historical faulty records with zero false positives.
- Root-caused a WER degradation of Parakeet-CTC-1.1B on Ascend NPU via layer-wise operator comparison (1,811 ops): an NPU-only branch in HuggingFace Transformers cast the float relative-position-bias mask to boolean, discarding positional information across 42 attention layers; fix verified on a 200-sample regression and reported upstream.
- Led a cross-platform accuracy-drift audit against GPU baselines, separating engineering artifacts from genuine platform drift across 8 outlier models and establishing comparability pre-checks (model ID, weights, inference engine, decoding parameters) for chip-accuracy statistics.

**Shenzhen Quant-Cloud Energy Network Technology Co., Ltd.**

Aug 2024

*R&D Intern, Artificial Intelligence Department**Shenzhen, China*

- Built an Anomaly Transformer-based bearing-temperature anomaly detector for wind-turbine gearbox predictive maintenance.

**Kenexs Intelligent Technology Co., Ltd.**

Jul 2023 – Sep 2023

*Software Development Intern**Shenzhen, China*

- Built a Qt log analyzer for industrial machine-vision cameras used by the test team.

**PRESENTATIONS**

**Oral:** SegWorld [1], EMNLP 2026 (upcoming, Nov 2026); Dynamic Spectral Clustering FL [4], WASA 2025 (Jun 2025, Tokyo).  
**Posters:** PEAM [2], Findings of EMNLP 2026 (upcoming, Nov 2026); FusionProxy [3], CVPR 2026 WMAS Workshop (Jun 2026).

**SKILLS****Programming:** Python, C/C++, CUDA, Java, SQL, JavaScript**ML & Systems:** PyTorch, HuggingFace Transformers, SGLang, NeMo, NumPy/SciPy, CUDA kernel optimization (Tensor Cores, cuBLAS), Linux, Git, LaTeX**Languages:** Chinese (native), English (fluent)